

TCS NQT

3 Hour test

2 parts : foundational + advanced

One exam decided your role

Ninja (3.5lpa), Digital(7lpa), Prime(9-10 LPA)

No negative marking

**Languages: C,C++,Java,Python,Perl

** AI proctored

** 60-80% of questions follow repeated patterns from past years

** Foundation cutoff decides if your Advanced even matters

Foundation:

Aptitude: 20 questions

Number System (LCM/HCF, divisibility), Percentages, Profit & Loss, Time & Work, Speed-Time-Distance, Averages, SI/CI, Ratio & Proportion, Permutations & Combinations, Probability, Mixtures & Allegations, Geometry

Verbal Ability: 25 questions

Reading Comprehension, Sentence Correction, Para Jumbles, Fill in the Blanks, Synonyms/Antonyms, Prepositions, Active/Passive Voice

Reasoning: 20 questions

Coding-Decoding, Blood Relations, Directions, Number Series, Data Sufficiency, Seating Arrangements, Syllogisms, Meaningful Word Formation

Advanced

Advanced Quant + Reasoning MCQs

DSA: 2 questions (med-little hard)

** partial credits

1st easy (story mode)

2nd hard

** edge cases

80% questions are:

second largest, reverse in-place, find duplicates, move zeros to end, rotate array, subarray sums

palindrome check, reverse, anagram, character frequency, substring problems

prime check (use **Sieve of Eratosthenes**), Fibonacci, factorial, Armstrong, GCD/LCM

star/number patterns with nested loops

bubble, insertion, binary search

20% (for prime and digital)

Recursion & Backtracking

Linked Lists (reverse, detect cycle, middle node)

Stacks (balanced parentheses, next greater element)

Trees (traversals, BFS/DFS)

Hashmaps (two sum, frequency counting)

Basic DP (Fibonacci variants, coin change)

Resources

Prepinsta TCS interface

GeeksforGeeks TCS NQT sheet

4 hours each day (3 for dsa, 1 for reasoning/apti)

For **dsa**: patterns Playlist. Just follow the video and questions

For **apti/reasoning**: paste this **prompt** in chatgpt/gemini/claude:

You are a TCS NQT aptitude/reasoning coach. Today's topic: [**TOPIC**].

1. Give me a 5-min concept refresher with key formulas and 2-3 shortcuts.
2. Then give me 10 questions of TCS NQT difficulty.
3. Don't show answers — I'll attempt first.
4. Mark my answers, show solutions for wrong ones with the fastest method.
5. End with 3 PYQ-style traps TCS commonly asks on this topic.

Day 1:

Arrays/DSA Basics — Traversal, find max/min, second largest, reverse in-place, count positive/negative/zeros

Basics:

choose your language and practice

<https://www.youtube.com/playlist?list=PLbJhGqY-mq44tdtsBxMG00DMttxm9JspQ>

Hashmaps:

https://www.youtube.com/watch?v=9a4A6CbrLCo&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=45

Apti

Number System: LCM, HCF, Divisibility, Remainders

Day 2:

2 pointers:

Move zeros to end:

https://www.youtube.com/watch?v=Fu7LD_mlo00&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=3

Remove Duplicates

https://www.youtube.com/watch?v=PvyEr3CeKzE&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=4

Triplet sum:

https://www.youtube.com/watch?v=rM9EthMIXnw&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=7

Dutch national flag:

https://www.youtube.com/watch?v=ljJcYql6Bc&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=8

Apti:

Percentages + basic shortcuts

Day 3:

Intro to sliding window

https://www.youtube.com/watch?v=V6pRTnOZ7Mc&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=10

Fixed window questions

https://www.youtube.com/watch?v=DL8LSXUsfWE&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=11

String window question:

https://www.youtube.com/watch?v=2HZ12B2ZPAQ&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=13

Apti:

Profit & Loss, Discounts

Day 4:

Sliding window revision

https://www.youtube.com/watch?v=lyZp-49pdzQ&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=14

Sliding window revision (min window)

https://www.youtube.com/watch?v=IR3oL6ltbJ8&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=15

Sliding window Imp question:

https://www.youtube.com/watch?v=9wc8HZH_sh4&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=16

Apti:

Time & Work, Pipes & Cisterns

Day 5:

Strings I — Reverse string, palindrome check, count vowels/consonants, character frequency

These are basic questions of 2 pointers. Use chatgpt for answers. You would have already learnt this if you are consistent

Apti:

Speed-Time-Distance, Trains, Boats

Day 6:

Number theory : Prime check (Sieve of erasthenes)

Armstrong, Perfect number, Factorial, Fibonacci

Use chatgpt to quickly learn these

Apti:

Ratio, Proportion, Mixtures & Allegations

Day 7:

Pattern Printing + Sorting — Star/number/pyramid patterns, Bubble sort, Insertion sort, Selection sort

Use chatgpt for these

Time complexity:

https://www.youtube.com/watch?v=oFwHwCkSoGw&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=6

Apti:

Averages, Ages, SI/CI

Day 8:

Linked List Basics

https://www.youtube.com/watch?v=NSh5oNEID84&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=20

Linked list Slow-fast intro

https://www.youtube.com/watch?v=lxslDYwMrZ8&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=17

Slow-fast question

https://www.youtube.com/watch?v=RNpZBhZBtJc&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=21

Apti:

P&C basics + Probability

Day 9:

Linked List reversal:

https://www.youtube.com/watch?v=7gOCx1vbx0k&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=47

Kadane Intro (max subarray)

https://www.youtube.com/watch?v=N8vJ8RyQEes&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=23

Kadane one deletion

https://www.youtube.com/watch?v=jlz4zcFuUU8&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=48

Apti

Geometry (areas, perimeters, basic mensuration)

Day 10:

Binary Search Intro:

https://www.youtube.com/watch?v=S0E1Ix67qbc&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=51

Binary Search Question

https://www.youtube.com/watch?v=zEEwwS9_fwA&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=52

Binary Search Story Questions

https://www.youtube.com/watch?v=0Kxg0LPGwFo&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=56

Reasoning

Coding-Decoding + Blood Relations

Day 11:

Stack intro

https://www.youtube.com/watch?v=V0wJuik3WSE&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=39

Stack Question

https://www.youtube.com/watch?v=K1idN2Rqcmw&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=40

Stack next greater

https://www.youtube.com/watch?v=gFqOIPfAEjw&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=42

Stack balanced parenthesis

https://www.youtube.com/watch?v=YutYOZFt6sQ&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=41

Reasoning: Directions + Number Series

Day 12:

Heap Intro

https://www.youtube.com/watch?v=_1AbrkD8pak&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=61

Heap Question

https://www.youtube.com/watch?v=zzu-k85RLGs&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=62

Heap on Pairs

https://www.youtube.com/watch?v=l1-0ALAJxZI&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=63

Heap Revision

https://www.youtube.com/watch?v=1yax0YQOb8w&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=69

Day 13:

Recursion Basics

https://www.youtube.com/watch?v=HdPb_thlF5s&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=70

Backtracking

https://www.youtube.com/watch?v=l081UkZCLlc&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=72

Recursion questions

https://www.youtube.com/watch?v=-gC-QEdpvO4&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=74

Backtracking template

https://www.youtube.com/watch?v=IKfIT6uFOcs&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=76

Reasoning: Syllogisms + Data Sufficiency

Day 14:

Tree intro

https://www.youtube.com/watch?v=VdujirLGDDE&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=79

Tree Traversal

https://www.youtube.com/watch?v=ZvYJT_QwnEw&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=80

Tree Level Order

https://www.youtube.com/watch?v=-g6h0Ok1Buk&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=81

Tree LCA

https://www.youtube.com/watch?v=_UoIHF3KUpE&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=84

Reasoning: Seating Arrangements + Meaningful Word Formation

Day 15:

Prefix Sum Intro

https://www.youtube.com/watch?v=F86WfZ5RUC8&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=29

Subarray sum 0

https://www.youtube.com/watch?v=XLtRoHIDGs0&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=30

Num of subarrays

https://www.youtube.com/watch?v=sh0Ng7sjscE&list=PLbJhGqY-mq47k_WLUtzVjmarUm1EuXPj2&index=31

Day 16:

1-d DP

Fibonacci

Coin change

Climbing stairs

{use any resource you like}

Day 17,18,19,20

Give mocks from prepinsta, identify the wrong questions specially in apti and you can brush up on verbal if you wish.